

- 11 27. A pendulum of length 75 cm and mass 2.5 kg swings with a frequency of 12 Hz.
- 12 28. A train sounds its whistle while passing by a railroad crossing. An observer at the crossing measures a frequency of 219 Hz as the train approaches the crossing and a frequency of 184 Hz as the train leaves. The speed of sound is 340 m/s. Find the speed of the train and the frequency of its whistle.
- 12 29. In order to decrease the fundamental frequency of a guitar string by 4.0 %, by what percentage should you reduce the tension?
- 12 30. If the velocity of sound in air at 27 °C and at a pressure of 76 cm of Hg is 345 m/s. Find the velocity at 127 °C and 75 cm of Hg.
- 13 31. A grating has exactly 8000 lined uniformly spaced over 2.54 cm and is illuminated by light from a mercury vapor discharge lamp. What is the expected angle for the third order maximum of the light of wavelength 546 nm?
- 13 32. A beam of X-rays of wavelength 0.071 nm is diffracted by a plane of rock salt with a distance between the atomic planes are 1.98 Å. Find the glancing angle for the second order diffraction.
- 13 33. A monochromatic light of wavelength 6900 Å is used to illuminate two parallel slits. On a screen that is 3.30 m away from the slits, interference fringes are observed. The distance between adjacent bright fringes in the center of pattern is 1.80 cm, what is the distance between the slits?

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Numerical no. 28;

Data;

$$f'_T = 219 \text{ Hz (towards)}$$

$$f'_A = 184 \text{ Hz (away)}$$

$$V = 340 \text{ ms}^{-1}$$

$$v_s = ?$$

$$f = ?$$

Solution;

Towards;

Away;

$$f'_{(T)} = \left(\frac{V}{V - v_s} \right) f \quad f'_{(A)} = \left(\frac{V}{V + v_s} \right) f$$

dividing both above equations;

$$\frac{f'_{(T)}}{f'_{(A)}} = \frac{\left(\frac{V}{V - v_s} \right) f}{\left(\frac{V}{V + v_s} \right) f}$$

$$\frac{f'_{(T)}}{f'_{(A)}} = \frac{(V - v_s)}{(V + v_s)}$$

$$\frac{219}{184} = \frac{340 - v_s}{340 + v_s}$$

$$1.19(340 + v_s) = 340 - v_s$$

$$1.404.6 + 1.19v_s = 340 - v_s$$

$$1.19v_s + v_s = 340 - 1.404.6$$

$$2.19v_s = -64.6$$

$$v_s = \frac{-64.6}{2.19}$$

$$v_s = 29.4 \text{ m/s} \quad \underline{\text{Ans!}}$$

∴ put the value of v_s in towards f' or in away f' eq the ans will be the same.

$$f'_{(T)} = \left(\frac{V}{V - v_s} \right) f$$

$$219 = \left(\frac{340}{340 - 29.4} \right) f$$

$$219 = 1.094f$$

$$\frac{219}{1.094} = f$$

$$f = 200 \text{ Hz} \quad \underline{\text{Ans!}}$$

Numeical no. 29;

Data;

$$f' = 100\% - 4\% = 96\% \text{ OR } 0.96f$$

$$\left(\frac{T-T'}{T}\right) \times 100 = ?$$

Solution;

$$f = \frac{1}{2\pi} \sqrt{\frac{T}{\mu}} \quad \text{--- (i)}$$

$$f' = \frac{1}{2\pi} \sqrt{\frac{T'}{\mu}} \quad \text{--- (ii)}$$

divide both eqs;

$$\frac{f}{f'} = \frac{\frac{1}{2\pi} \sqrt{\frac{T}{\mu}}}{\frac{1}{2\pi} \sqrt{\frac{T'}{\mu}}}$$

$$\frac{f}{0.96f} = \sqrt{\frac{T}{T'}}$$

$$0.96 = \sqrt{\frac{T'}{T}}$$

squaring on both sides

$$(0.96)^2 = \left(\sqrt{\frac{T'}{T}}\right)^2$$

$$0.92 = \frac{T'}{T}$$

$$T' = 0.92T$$

$$\therefore = \left(\frac{T-T'}{T}\right) \times 100$$

$$\therefore = \frac{T - 0.92T}{T} \times 100$$

$$\therefore = \frac{T(1-0.92)}{T} \times 100$$

$$\therefore = 8\% \quad \text{Ans!}$$

Numerical no. 30

Data;

$$T_1 = 27^\circ\text{C} + 273 = 300^\circ\text{K}$$

$$V_1 = 345 \text{ m/s}$$

$$T_2 = 127^\circ\text{C} + 273 = 400^\circ\text{K}$$

$$V_2 = ?$$

Solution;

$$\frac{V_1}{V_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\frac{V_1}{V_2} = \sqrt{\frac{300}{400}}$$

$$\frac{V_1}{V_2} = 0.866$$

$$V_2 = \frac{V_1}{0.866}$$

$$V_2 = \frac{345}{0.866}$$

$$V_2 = 398.38 \text{ m/s} \quad \underline{\text{Ans!}}$$

Numerical no. 31

Data;

$$N = 8000$$

$$L = 2.54 \text{ cm} = 2.54 \times 10^{-2} \text{ m}$$

$$\lambda = 546 \text{ nm} = 546 \times 10^{-9} \text{ m}$$

$$m = 3$$

$$\theta = ?$$

Solution;

$$d = \frac{L}{N}$$

$$d = \frac{2.54 \times 10^{-2}}{8000}$$

$$d = 3.175 \times 10^{-6} \text{ m}$$

$$m \lambda = d \sin \theta$$

$$\theta = \sin^{-1} \left(\frac{m \lambda}{d} \right)$$

$$\theta = \sin^{-1} \left(\frac{3(546 \times 10^{-9})}{3.175 \times 10^{-6}} \right)$$

$$\theta = 31.05^\circ \text{ Ans!}$$

Numerical no. 32

Data;

$$\lambda = 0.071 \text{ nm} = 0.071 \times 10^{-9} \text{ m}$$

$$m = 2$$

$$d = 1.98 \text{ \AA} = 1.98 \times 10^{-10} \text{ m}$$

$$\theta = ?$$

Solution:-

$$m \lambda = d \sin \theta$$

$$\theta = \sin^{-1} \left(\frac{m \lambda}{d} \right)$$

$$\theta = \sin^{-1} \left(\frac{2(0.071 \times 10^{-9})}{2(1.98 \times 10^{-10})} \right)$$

$$\theta = 0.35^\circ \text{ Ans!}$$

Numerical no. 33

Data;

$$\lambda = 6900 \text{ \AA} = 6900 \times 10^{-10} \text{ m}$$

$$L = 3.30 \text{ m}$$

$$\Delta x = 1.80 \text{ cm} = 1.80 \times 10^{-2} \text{ m}$$

$$d = ?$$

Solution;

$$\Delta x = \frac{\lambda L}{d}$$

$$d = \frac{\lambda L}{\Delta x}$$

$$d = \frac{6900 \times 10^{-10} (3.30)}{1.80 \times 10^{-2}}$$

$$d = 1.265 \times 10^{-4} \text{ m} \quad \text{Ans!}$$